

Computer Simulation Problems

SIM Problems identified by the Multisim/PSpice icon are intended to demonstrate the value of using SPICE simulation to verify hand analysis and design, and to investigate important issues such as allowable signal swing and amplifier nonlinear distortion. Instructions to assist in setting up PSpice and Multisim simulations for all the indicated problems can be found in the corresponding files on the website. Note that if a particular parameter value is not specified in the problem statement, you are to make a reasonable assumption.

Section 9.1: The MOS Differential Pair

9.1 For an NMOS differential pair with a common-mode voltage V_{CM} applied, as shown in Fig. 9.2, let $V_{DD} = V_{SS} = 1.0$ V, $k'_n = 0.4$ mA/V², $(W/L)_{1,2} = 10$, $V_{tn} = 0.4$ V, $I = 0.16$ mA, $R_D = 5$ k Ω , and neglect channel-length modulation.

- Find V_{OV} and V_{GS} for each transistor.
- For $V_{CM} = 0$, find V_S , I_{D1} , I_{D2} , V_{D1} , and V_{D2} .
- Repeat (b) for $V_{CM} = +0.4$ V.
- Repeat (b) for $V_{CM} = -0.1$ V.
- What is the highest value of V_{CM} for which Q_1 and Q_2 remain in saturation?
- If current source I requires a minimum voltage of 0.2 V to operate properly, what is the lowest value allowed for V_S and hence for V_{CM} ?

9.2 For the PMOS differential amplifier shown in Fig. P9.2 let $V_{tp} = -0.8$ V and $k'_p W/L = 4$ mA/V². Neglect channel-length modulation.

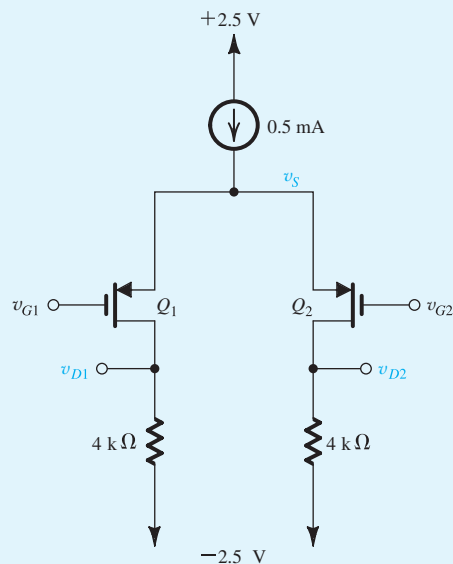


Figure P9.2

- For $v_{G1} = v_{G2} = 0$ V, find $|V_{OV}|$ and V_{SG} for each of Q_1 and Q_2 . Also find V_S , V_{D1} , and V_{D2} .
- If the current source requires a minimum voltage of 0.4 V, find the input common-mode range.

9.3 For the differential amplifier specified in Problem 9.1 let $v_{G2} = 0$ and $v_{G1} = v_{id}$. Find the value of v_{id} that corresponds to each of the following situations:

- $i_{D1} = i_{D2} = 0.08$ mA; (b) $i_{D1} = 0.12$ mA and $i_{D2} = 0.04$ mA;
- $i_{D1} = 0.16$ mA and $i_{D2} = 0$ (Q_2 just cuts off); (d) $i_{D1} = 0.04$ mA and $i_{D2} = 0.12$ mA; (e) $i_{D1} = 0$ mA (Q_1 just cuts off) and $i_{D2} = 0.16$ mA. For each case, find v_S , v_{D1} , v_{D2} , and $(v_{D2} - v_{D1})$.

SIM 9.4 For the differential amplifier specified in Problem 9.2, let $v_{G2} = 0$ and $v_{G1} = v_{id}$. Find the range of v_{id} needed to steer the bias current from one side of the pair to the other. At each end of this range, give the value of the voltage at the common-source terminal and the drain voltages.

9.5 Consider the differential amplifier specified in Problem 9.1 with G_2 grounded and $v_{G1} = v_{id}$. Let v_{id} be adjusted to the value that causes $i_{D1} = 0.09$ mA and $i_{D2} = 0.07$ mA. Find the corresponding values of v_{GS2} , v_S , v_{GS1} , and hence v_{id} . What is the difference output voltage $v_{D2} - v_{D1}$? What is the voltage gain $(v_{D2} - v_{D1})/v_{id}$? What value of v_{id} results in $i_{D1} = 0.07$ mA and $i_{D2} = 0.09$ mA?

D 9.6 Design the circuit in Fig. P9.6 to obtain a dc voltage of +0.1 V at each of the drains of Q_1 and Q_2 when $v_{G1} = v_{G2} = 0$ V. Operate all transistors at $V_{OV} = 0.15$ V

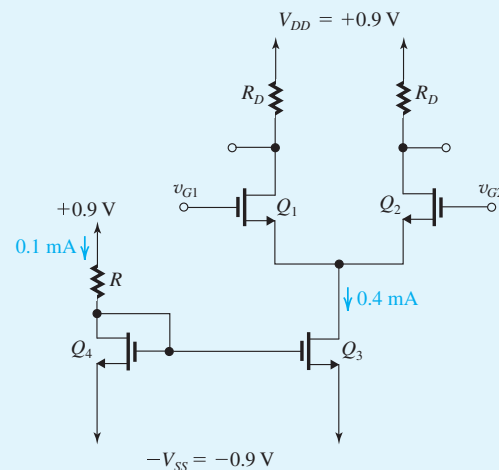


Figure P9.6

and assume that for the process technology in which the circuit is fabricated, $V_m = 0.4$ V and $\mu_n C_{ox} = 400 \mu\text{A/V}^2$. Neglect channel-length modulation. Determine the values of R , R_D , and the W/L ratios of Q_1 , Q_2 , Q_3 , and Q_4 . What is the input common-mode voltage range for your design?

9.7 The table providing the answers to Exercise 9.3 shows that as the maximum input signal to be applied to the differential pair is increased, linearity is maintained at the same level by operating at a higher V_{OV} . If $|v_{id}|_{\max}$ is to be 220 mV, use the data in the table to determine the required V_{OV} and the corresponding values of W/L and g_m .

9.8 Use Eq. (9.23) to show that if the term involving v_{id}^2 is to be kept to a maximum value of k then the maximum possible fractional change in the transistor current is given by

$$\frac{\Delta I_{\max}}{I/2} = 2\sqrt{k(1-k)}$$

and the corresponding maximum value of v_{id} is given by

$$v_{id\max} = 2\sqrt{k}V_{OV}$$

Evaluate both expressions for $k = 0.01$, 0.1 , and 0.2 .

9.9 A MOS differential amplifier biased with a current source $I = 200 \mu\text{A}$ is found to switch currents completely to one side of the pair when a difference signal $v_{id} = 0.3$ V is applied. At what overdrive voltage will each of Q_1 and Q_2 be operating when $v_{id} = 0$? If v_{id} for full current switching is to be 0.5 V, what must the bias current I be changed to?

D 9.10 Design the MOS differential amplifier of Fig. 9.5 to operate at $V_{OV} = 0.25$ V and to provide a transconductance g_m of 1 mA/V. Specify the W/L ratios and the bias current. The technology available provides $V_t = 0.5$ V and $\mu_n C_{ox} = 400 \mu\text{A/V}^2$.

9.11 For the MOS differential pair in Fig. 9.5, specify the value of $v_{id} \equiv v_{G1} - v_{G2}$, in terms of V_{OV} , that

- causes i_{D1} to increase by 10% above its equilibrium value of $I/2$.
- makes $i_{D1}/i_{D2} = 1.0; 2.0; 1.1; 1.01; 20$.

9.12 An NMOS differential amplifier is operated at a bias current I of 0.2 mA and has a W/L ratio of 32 , $\mu_n C_{ox} = 200 \mu\text{A/V}^2$, $V_A = 10$ V, and $R_D = 10$ k Ω . Find V_{OV} , g_m , r_o , and A_d .

D 9.13 It is required to design an NMOS differential amplifier to operate with a differential input voltage that

can be as high as 0.1 V while keeping the nonlinear term under the square root in Eq. (9.23) to a maximum of 0.04 . A transconductance g_m of 2 mA/V is needed and the amplifier is required to provide a differential output signal of 1 V when the input is at its maximum value. Find the required values of V_{OV} , I , R_D , and W/L . Assume that the technology available has $\mu_n C_{ox} = 200 \mu\text{A/V}^2$ and $\lambda = 0$.

D 9.14 Design a MOS differential amplifier to operate from ± 1 -V power supplies and dissipate no more than 1 mW in the equilibrium state. The differential voltage gain A_d is to be 10 V/V and the output common-mode dc voltage is to be 0.2 V. (Note: This is the dc voltage at the drains.) Assume $\mu_n C_{ox} = 400 \mu\text{A/V}^2$ and neglect the Early effect. Specify I , R_D , and W/L .

D 9.15 Design a MOS differential amplifier to operate from ± 1 -V supplies and dissipate no more than 1 mW in its equilibrium state. Select the value of V_{OV} so that the value of v_{id} that steers the current from one side of the pair to the other is 0.25 V. The differential voltage gain A_d is to be 10 V/V. Assume $k'_n = 400 \mu\text{A/V}^2$ and neglect the Early effect. Specify the required values of I , R_D , and W/L .

9.16 An NMOS differential amplifier employing equal drain resistors, $R_D = 47$ k Ω , has a differential gain A_d of 20 V/V.

- What is the value of g_m for each of the two transistors?
- If each of the two transistors is operating at an overdrive voltage $V_{OV} = 0.2$ V, what must the value of I be?
- For $v_{id} = 0$, what is the dc voltage across each R_D ?
- If v_{id} is 20 -mV peak-to-peak sine wave applied in a balanced manner but superimposed on $V_{CM} = 0.5$ V, what is the peak of the sine-wave signal at each drain?
- What is the lowest value that V_{DD} must have to ensure saturation-mode operation for Q_1 and Q_2 at all times? Assume $V_t = 0.5$ V.

9.17 A MOS differential amplifier is designed to have a differential gain A_d equal to the voltage gain obtained from a common-source amplifier. Both amplifiers utilize the same values of R_D and supply voltages, and all the transistors have the same W/L ratios. What must the bias current I of the differential pair be relative to the bias current I_D of the CS amplifier? What is the ratio of the power dissipation of the two circuits?

9.18 A differential amplifier is designed to have a differential voltage gain equal to the voltage gain of a common-source amplifier. Both amplifiers use the same values of R_D and supply voltages and are designed to dissipate equal amounts of power in their equilibrium or quiescent state. As well, all the transistors use the same channel length. What must the width W of the differential-pair transistors be relative to the width of the CS transistor?

D 9.19 Figure P9.19 shows a MOS differential amplifier with the drain resistors R_D implemented using diode-connected PMOS transistors, Q_3 and Q_4 . Let Q_1 and Q_2 be matched, and Q_3 and Q_4 be matched.

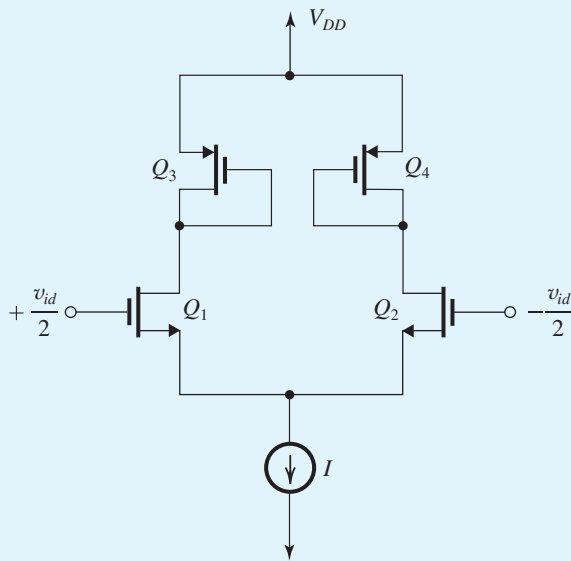


Figure P9.19

- Find the differential half-circuit and use it to derive an expression for A_d in terms of $g_{m1,2}$, $g_{m3,4}$, $r_{o1,2}$, and $r_{o3,4}$.
- Neglecting the effect of the output resistances r_o , find A_d in terms of μ_n , μ_p , $(W/L)_{1,2}$ and $(W/L)_{3,4}$.
- If $\mu_n = 4\mu_p$ and all four transistors have the same channel length, find $(W_{1,2}/W_{3,4})$ that results in $A_d = 10$ V/V.

9.20 Find the differential half-circuit for the differential amplifier shown in Fig. P9.20 and use it to derive an expression for the differential gain $A_d \equiv v_{od}/v_{id}$ in terms of g_m , R_D , and R_s . Neglect the Early effect. What is the gain with

$R_s = 0$? What is the value of R_s (in terms of $1/g_m$) that reduces the gain to half this value?

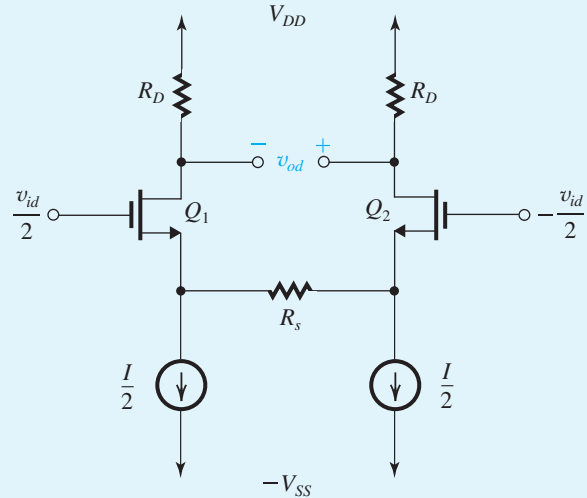


Figure P9.20

***9.21** The resistance R_s in the circuit of Fig. P9.20 can be implemented by using a MOSFET operated in the triode region, as shown in Fig. P9.21. Here Q_3 implements R_s , with the value of R_s determined by the voltage V_c at the gate of Q_3 .

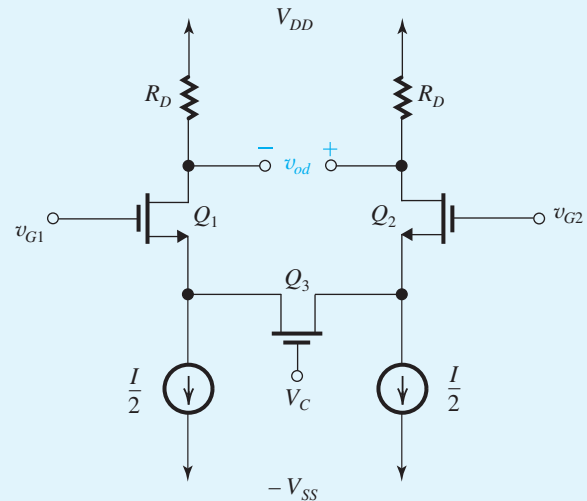


Figure P9.21

- With $v_{G1} = v_{G2} = 0$ V, and assuming that Q_1 and Q_2 are operating in saturation, what dc voltages appear